공학수학(2)

14.8 Triple Integrals in Spherical Coordinates

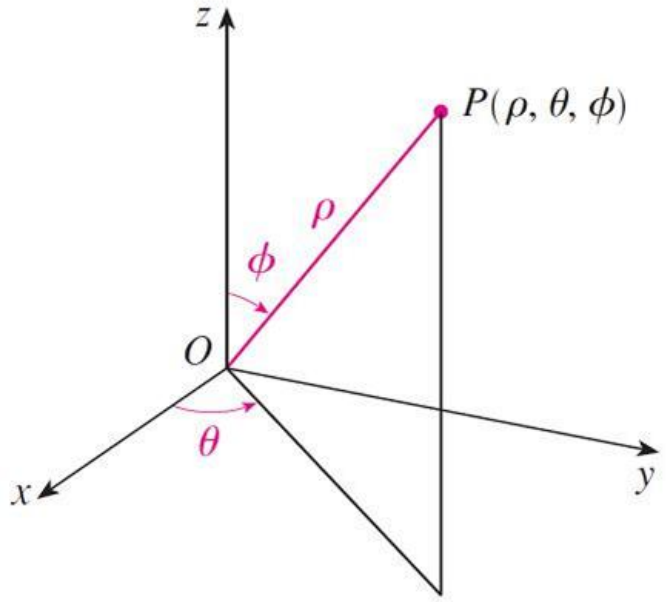
Triple Integrals in Spherical Coordinates

Another useful coordinate system in three dimensions is the *spherical coordinate system*.

It simplifies the evaluation of triple integrals over regions bounded by spheres or cones.

Spherical Coordinates

The **spherical coordinates** (ρ, θ, ϕ) of a point P in space are shown in Figure 1 where $\rho = |OP|$ is the distance from the origin to P, θ is the same angle as in cylindrical coordinates, and ϕ is the angle between the positive z-axis and the line segment OP.



The spherical coordinates of a point

Figure 1

Spherical Coordinates

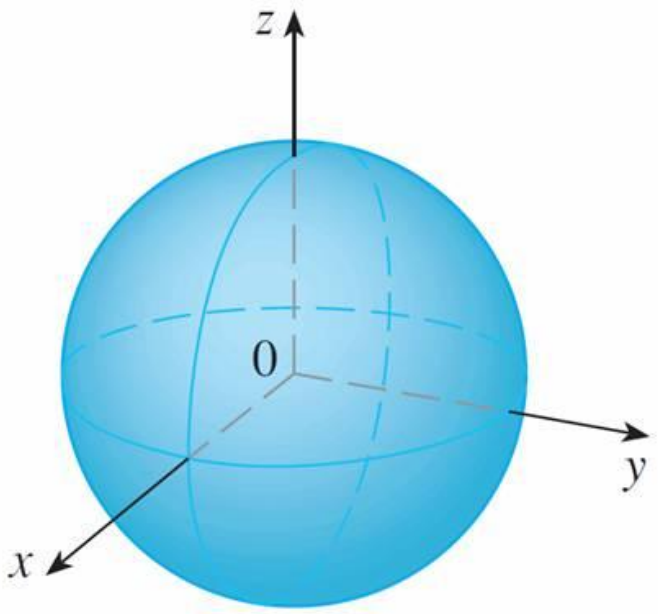
Note that

$$\rho \geq 0 \qquad 0 \leq \phi \leq \pi$$

The spherical coordinate system is especially useful in problems where there is symmetry about a point, and the origin is placed at this point.

Spherical Coordinates

For example, the sphere with center the origin and radius c has the simple equation $\rho = c$ (see Figure 2); this is the reason for the name “spherical” coordinates.

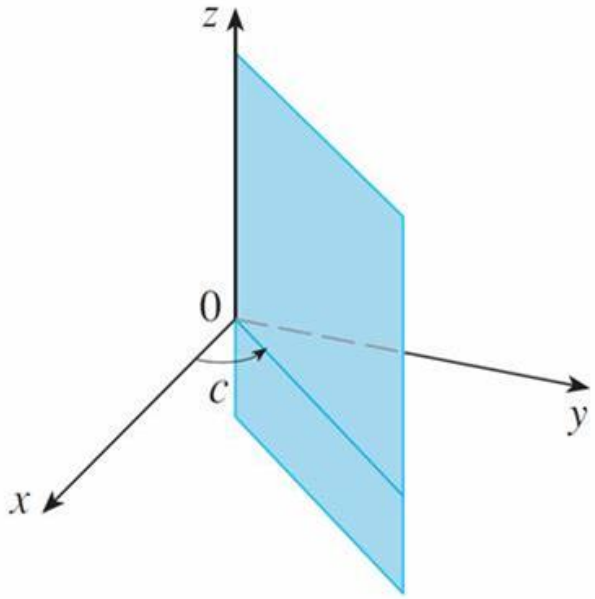


$\rho = c$, a sphere

Figure 2

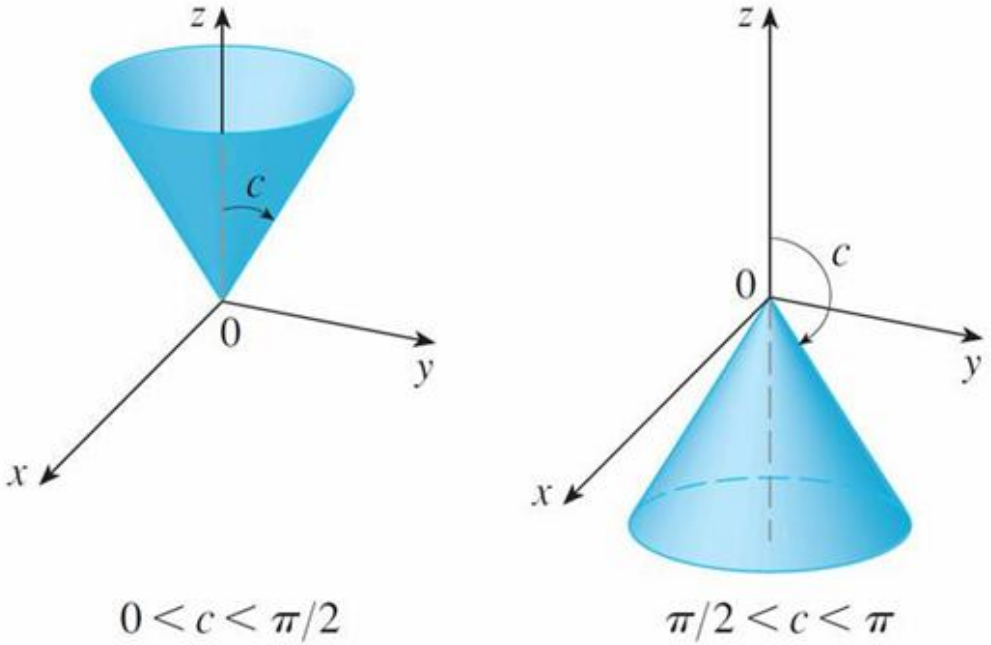
Spherical Coordinates

The graph of the equation $\theta = c$ is a vertical half-plane (see Figure 3), and the equation $\phi = c$ represents a half-cone with the z-axis as its axis (see Figure 4).



$\theta = c$, a half-plane

Figure 3



$\phi = c$, a half-cone

Figure 4

Spherical Coordinates

The relationship between rectangular and spherical coordinates can be seen from Figure 5.

From triangles OPQ and OPP' we have

$$z = \rho\cos\phi \qquad r = \rho\sin\phi$$

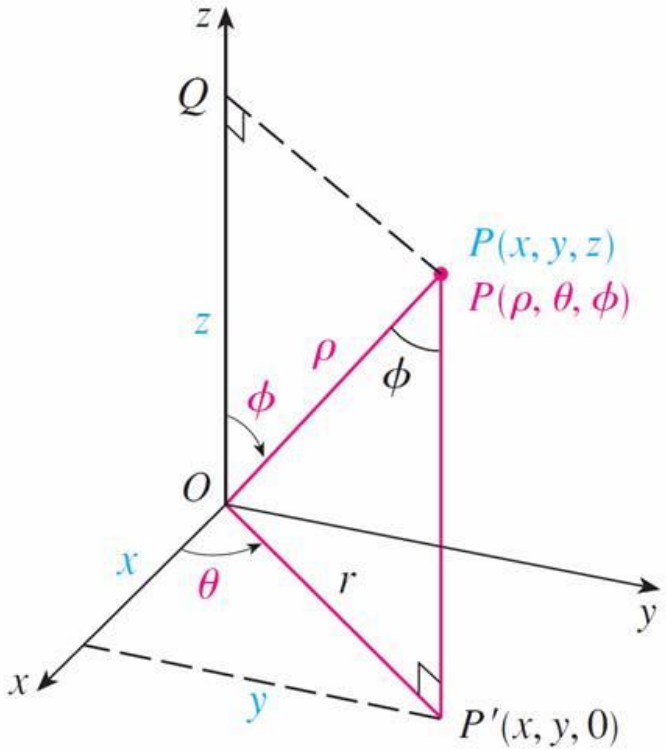


Figure 5

Spherical Coordinates

But $x = r\cos\theta$ and $y = r\sin\theta$, so to convert from spherical to rectangular coordinates, we use the equations

$$\textbf{1} \qquad x = \rho\sin\phi\cos\theta \qquad y = \rho\sin\phi\sin\theta \qquad z = \rho\cos\phi$$

Also, the distance formula shows that

$$\textbf{2} \qquad \rho^2 = x^2 + y^2 + z^2$$

We use this equation in converting from rectangular to spherical coordinates.

Example 1

The point $(2,\ \pi/4,\ \pi/3)$ is given in spherical coordinates. Plot the point and find its rectangular coordinates.

Solution:

We plot the point in Figure 6.

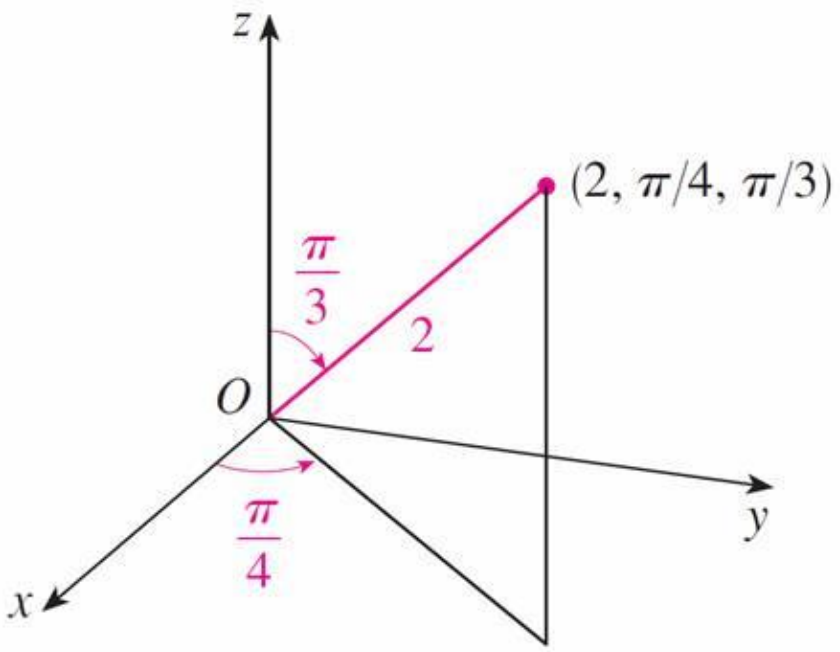


Figure 6

Triple Integrals in Spherical Coordinates

In the spherical coordinate system the counterpart of a rectangular box is a **spherical wedge**

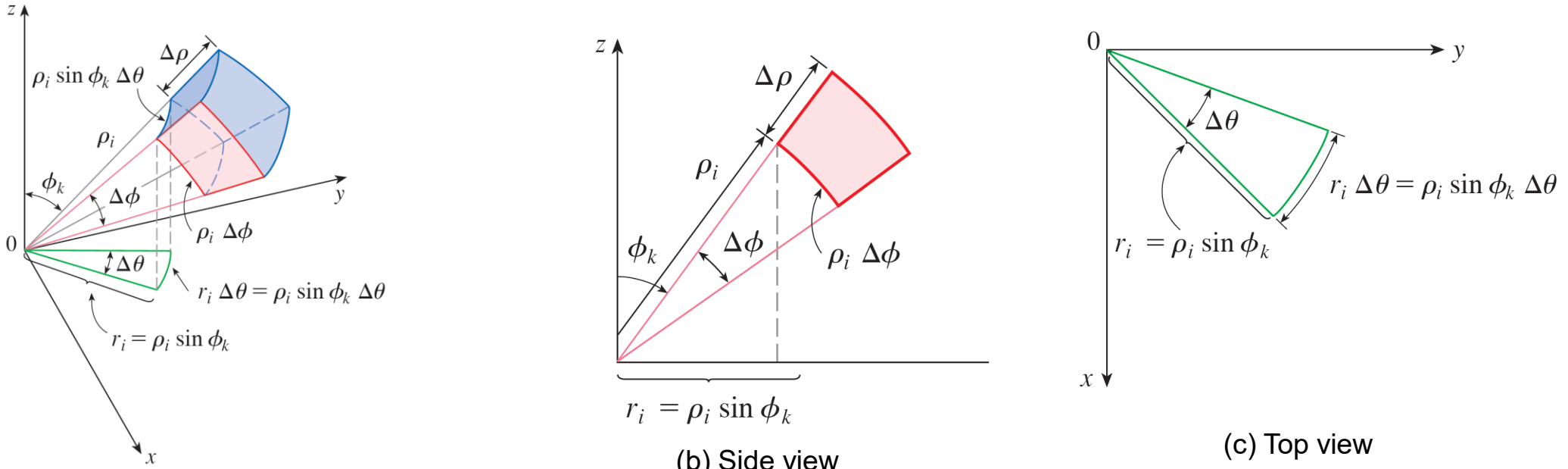
$$E = \{(\rho,\ \theta,\ \phi) \mid a \le \rho \le b,\ \alpha \le \theta \le \beta,\ c \le \phi \le d\}$$

where $a \ge 0$ and $\beta - \alpha \le 2\pi$, and $d - c \le \pi$. Although we defined triple integrals by dividing solids into small boxes, it can be shown that dividing a solid into small spherical wedges always gives the same result.

So we divide E into smaller spherical wedges E_{ijk} by means of equally spaced spheres $\rho = \rho_i$, half-planes $\theta = \theta_j$, and half-cones $\phi = \phi_k$.

Triple Integrals in Spherical Coordinates

Figure 7 shows that E_{ijk} is approximately a rectangular box with dimensions $\Delta\rho$, $\rho_i\ \Delta\phi$ (arc of a circle with radius ρ_i, angle $\Delta\phi$), and $\rho_i\ \sin\phi_k\ \Delta\theta$ (arc of a circle with radius $\rho_i\ \sin\phi_k$, angle $\Delta\theta$).



(a) A spherical wedge

(b) Side view

(c) Top view

Figure 7

Triple Integrals in Spherical Coordinates

So an approximation to the volume of E_{ijk} is given by

$$\Delta V_{ijk} \approx (\Delta\rho)(\rho_i \Delta\phi)(\rho_i \sin\phi_k \Delta\theta) = \rho_i^2 \sin\phi_k \Delta\rho\, \Delta\theta\, \Delta\phi$$

In fact, it can be shown, with the aid of the Mean Value Theorem, that the volume of E_{ijk} is given exactly by

$$\Delta V_{ijk} = \tilde{\rho}_i^2 \sin\tilde{\phi}_k \, \Delta\rho\, \Delta\theta\, \Delta\phi$$

where $\left(\rho_i,\ \theta_j,\ \phi_k\right)$ is some point in E_{ijk}.

Triple Integrals in Spherical Coordinates

Let $\left(x_{ijk}^*,\ y_{ijk}^*,\ z_{ijk}^*\right)$ be the rectangular coordinates of this point.

Then

$$\iiint_F f(x,\ y,\ z)\,dV = \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f\left(x_{ijk}^*,\ y_{ijk}^*,\ z_{ijk}^*\right)\Delta V_{ijk}$$

$$= \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f\left(\rho_i \sin\phi_k \cos\theta_j,\ \rho_i \sin\phi_k \sin\theta_j,\ \rho_i \cos\phi_k\right)\rho_i^2 \sin\phi_k \Delta\rho\ \Delta\theta\ \Delta\phi$$

Triple Integrals in Spherical Coordinates

But this sum is a Riemann sum for the function

$$F(\rho,\ \theta,\ \phi) = f(\rho\sin\phi\cos\theta,\ \rho\sin\phi\sin\theta,\ \rho\cos\phi)\,\rho^2\ \sin\phi$$

Consequently, we have arrived at the following **formula for triple integration in spherical coordinates**.

3
$$\iiint_E f(x,\ y,\ z)\ dv$$
$$= \int_c^d \int_\alpha^\beta \int_a^b f(\rho\sin\phi\cos\theta,\ \rho\sin\phi\sin\theta,\ \rho\cos\phi)\,\rho^2\sin\phi\ d\rho\ d\theta\ d\phi$$

where E is a spherical wedge given by

$$E = \{(\rho,\ \theta,\ \phi)\,|\,a \le \rho \le b,\ \alpha \le \theta \le \beta,\ c \le \phi \le d\}$$

Triple Integrals in Spherical Coordinates

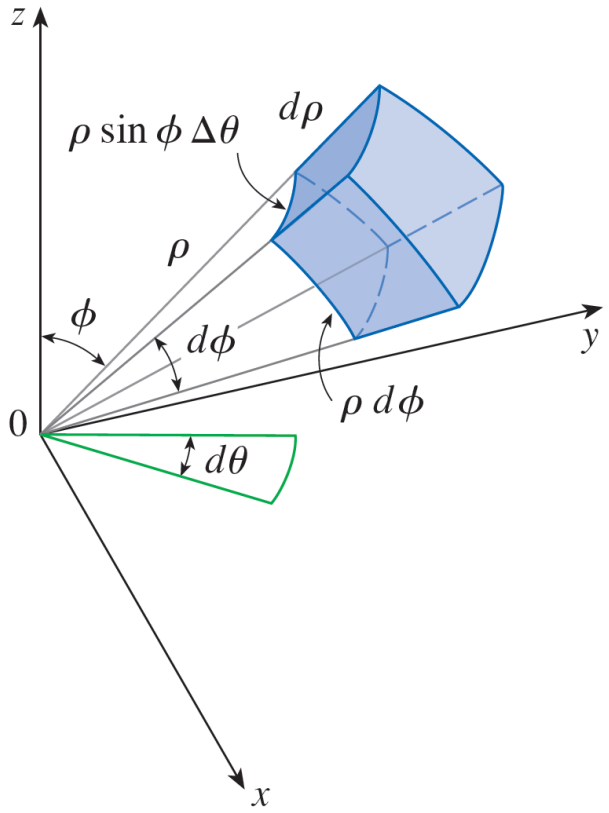
Formula 3 says that we convert a triple integral from rectangular coordinates to spherical coordinates by writing

$$x = \rho \sin\phi \cos\theta \qquad y = \rho \sin\phi \sin\theta \qquad z = \rho \cos\phi$$

using the appropriate limits of integration, and replacing dv by $\rho^2 \sin\phi \; d\rho \; d\theta \; d\phi$.

Triple Integrals in Spherical Coordinates

This is illustrated in Figure 8.



Volume element in spherical coordinates: $dv = \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$.

Figure 8

Triple Integrals in Spherical Coordinates

This formula can be extended to include more general spherical regions such as

$$E = \left\{ \left(\rho,\ \theta,\ \phi \right) \mid \alpha \le \theta \le \beta,\ c \le \phi \le d,\ g_1\left(\theta,\ \phi \right) \le \rho \le g_2\left(\theta,\ \phi \right) \right\}$$

In this case the formula is the same as in (3) except that the limits of integration for ρ are $g_1(\theta, \phi)$ and $g_2(\theta, \phi)$.

Usually, spherical coordinates are used in triple integrals when surfaces such as cones and spheres form the boundary of the region of integration.

Example 4

Use spherical coordinates to find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$. (See Figure 9.)

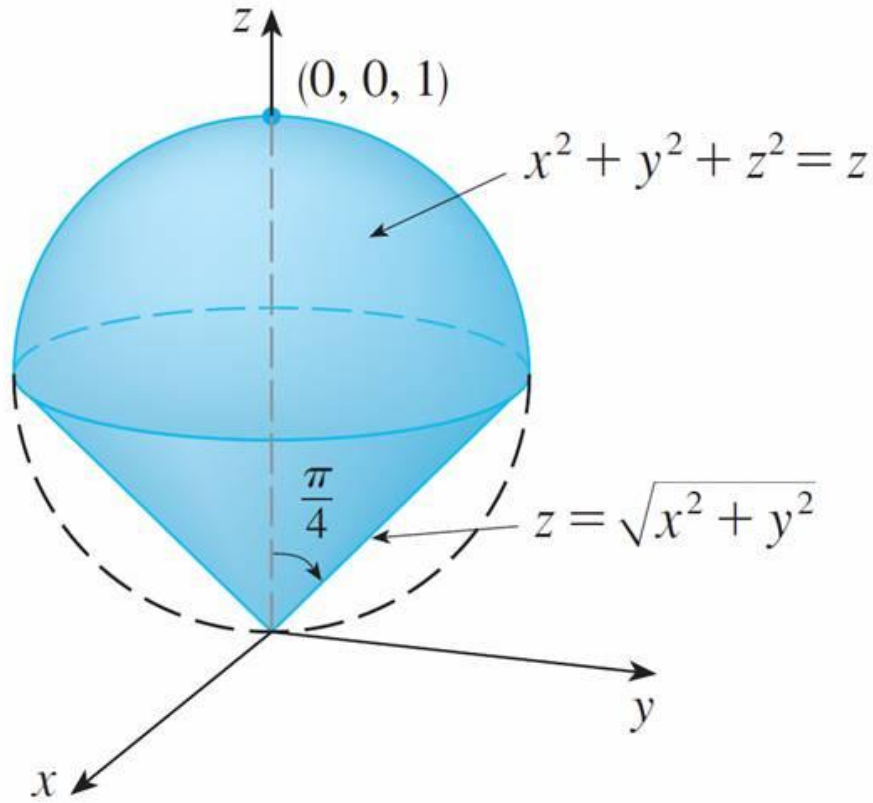
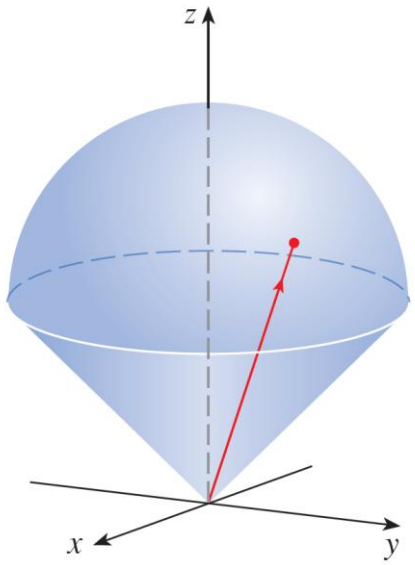


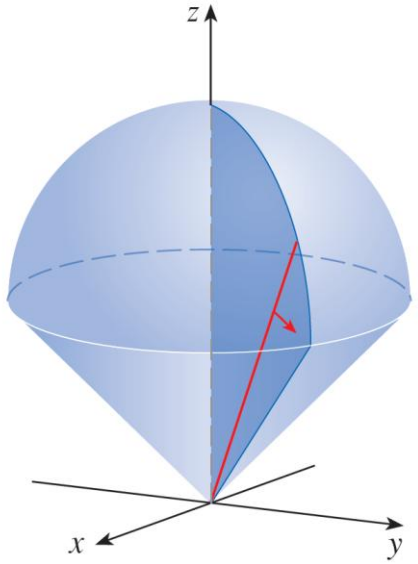
Figure 9

Example 4

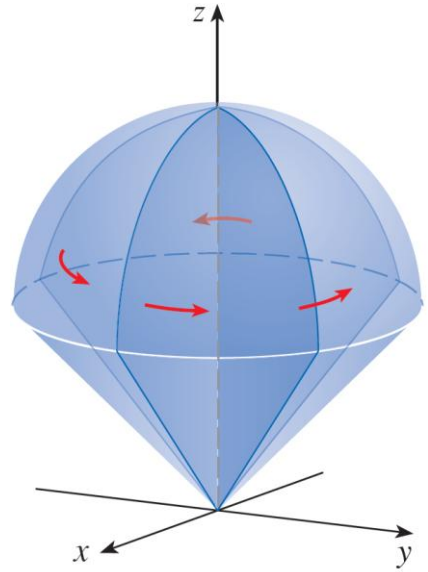
Figure 10 shows how E is swept out if we integrate first with respect to ρ, then ϕ, and then θ.



ρ varies from 0 to $\cos\phi$ while ϕ and θ are constant.



ϕ varies from 0 to $\pi/4$ while θ is constant.



θ varies from 0 to 2π.

Figure 10